Lesson 4: Data Storage & Compression - My work

Name: Test Student Date: 15 October 2026

Recap - Lesson 3

1. How many different colours can be shown with a colour depth of 6 bits?

Answer: 64 [Correct]

2. An image is 120 x 100 pixels with a colour depth of 8 bits. What is its file size in bytes?

Answer: 12,000 bytes [Correct]

3. A 4-second sound is recorded at 22,050 Hz with a sample resolution of 8 bits. What is its file size in bytes?

Answer: 88200 [Correct]

Levels 1-4: Getting started

Your turn

1. How many bits are in 1 byte?

Answer: 8 [Correct]

2. How many bits are in 1 nibble?

Answer: 4 [Correct]

3. How many bytes are in 1 KiB?

Answer: 1024 [Correct]

4. Convert 5 KiB to bytes.

Answer: 5120 [Correct]

5. Convert 2048 KiB to MiB.

Answer: 2 [Correct]

6. Use RLE to encode AAAABBB.

Answer: 4a 3b [Correct]

Quiz - score 8 / 8

Q1. Which of these is the largest unit?

Answer: GiB [Correct]

Q2. How many KiB are in 1 MiB?

Answer: 1024 [Correct]

Q3. Convert 24 bits to bytes.

Answer: 3 [Correct]

Q4. Convert 3072 bytes to KiB.

Answer: 3 [Correct]

Q5. What is data compression?

Answer: Making a file's size smaller [Correct]

Q6. Which type of compression permanently removes data?

Answer: Lossy [Correct]

Q7. After lossless compression, can the original file be restored exactly?

Answer: Yes [Correct]

Q8. Use RLE to encode CCCCCDD.

Answer: 5C2D [Correct]

Exam-style question

Give two reasons why a file might be compressed before it is sent by email. [2]

(not answered)

Levels 4-7: Getting there

Your turn

1. Convert 6 MiB to KiB.

Answer: 6144 [Correct]

2. Convert 2.5 GiB to MiB.

Answer: 2560 [Correct]

3. Convert 98,304 bytes to KiB.

Answer: 96 [Correct]

4. An image is 800 x 600 pixels with a 16-bit colour depth. What is its size in KiB?

Answer: 937.5 [Correct]

5. A 2-minute mono recording uses 8,000 Hz and 16-bit sample resolution. What is its size in KiB?

Answer: 1875 [Correct]

6. Use RLE to encode the 1-bit image row WWWWWWBBWWWB.

Answer: 6W2B3W1B [Correct]

Quiz - score 8 / 8

Q1. Convert 12,288 KiB to MiB.

Answer: 12 [Correct]

Q2. An image is 512 x 512 pixels with an 8-bit colour depth. What is its size in KiB?

Answer: 256 [Correct]

Q3. A 10-second mono sound uses 32,000 Hz and 8-bit sample resolution. What is its size in KiB?

Answer: 312.5 [Correct]

Q4. Which compression would be most suitable for a file of program code?

Answer: Lossless, because every character must be restored exactly [Correct]

Q5. Which of these is a lossy method of reducing a music file's size?

Answer: Removing sounds that humans cannot hear [Correct]

Q6. Decode the RLE data 3R2G4B. (Type the full uncompressed text.)

Answer: RRRGGBBBB [Correct]

Q7. Why might a video be compressed before being uploaded to a website?

Answer: So it uploads faster and uses less storage and bandwidth [Correct]

Q8. How many characters does the RLE version of AAAAAAAABBBB use?

Answer: 4 [Correct]

Exam-style question

Explain the difference between lossy and lossless compression. For each, give a type of file it would be suitable for. [4]

Lossy removes data permanently, e.g. music; lossless keeps all data, e.g. text files.

Levels 7-9: Stretch

Your turn

1. An image is 1920 x 1080 pixels with a 24-bit colour depth. What is its size in MiB, to 2 decimal places?

Answer: 5.93 [Correct]

2. A 3-minute stereo recording uses 44,100 Hz and 16-bit sample resolution. What is its size in MiB, to 2 decimal places?

Answer: 30.28 [Correct]

3. How many whole 6 MiB photos can be stored on a 4 GiB memory card?

Answer: 682 [Correct]

4. An image file is exactly 2 MiB. It is 1024 x 512 pixels. What colour depth (in bits) was used?

Answer: 32 [Correct]

5. A mono sound file is 2400 KiB. It was recorded at 16,000 Hz with 16-bit sample resolution. How long is it, in seconds?

Answer: 76.8 [Correct]

6. AAAAABAAAAA is encoded with RLE as 5A1B5A. By what percentage has the number of characters been reduced? Give your answer to 1 decimal place.

Answer: 45.5 [Correct]

Quiz - score 8 / 8

Q1. A 1024 x 1024 image with a 24-bit colour depth is compressed to 25% of its original size. What is the compressed size in KiB?

Answer: 768 [Correct]

Q2. A 10-second stereo recording uses 48,000 Hz and 24-bit sample resolution. What is its size in MiB, to 2 decimal places?

Answer: 2.75 [Correct]

Q3. A sound file is 4 MiB. The sample resolution is reduced from 16 bits to 8 bits (nothing else changes). What is the new size in MiB?

Answer: 2 [Correct]

Q4. Why is RLE usually a poor choice for compressing a photograph?

Answer: Neighbouring pixels are rarely identical, so runs are short and the file may even get bigger [Correct]

Q5. A JPEG photo is edited and re-saved with lossy compression several times. Which statement is correct?

Answer: Quality can get worse each time, because removed data cannot be recovered [Correct]

Q6. How many whole 3.5 MiB songs fit on a 2 GiB music player?

Answer: 585 [Correct]

Q7. How many characters does the RLE version of ABABABAB use?

Answer: 16 [Correct]

Q8. A 640 x 480 image has its colour depth reduced from 24 bits to 8 bits. How many KiB are saved?

Answer: 600 [Correct]

Exam-style question

A music streaming company wants to reduce the size of the song files it sends to listeners. Evaluate the use of lossy and lossless compression for this purpose, and justify which you would recommend. [6]

(not answered)